

PilgrimIQ : Personalized Pilgrimage Decision Support System

ABSTRACT

PilgrimIQ : Personalized Pilgrimage Decision Support System is a web-based platform designed to improve the safety, health, and overall pilgrimage experience through intelligent decision support and personalized travel assistance. Pilgrimage journeys often involve long travel durations, crowded environments, physically demanding routes, unpredictable weather conditions, and health-related risks, especially for elderly pilgrims and individuals with pre-existing medical conditions. The platform enables pilgrims to create secure profiles, manage health records, upload medical reports, and plan their journeys using Artificial Intelligence (AI), Machine Learning (ML), Optical Character Recognition (OCR), and Natural Language Processing (NLP).

The system analyzes medical reports, generates health summaries, and classifies pilgrims into Low, Moderate, High, or Critical risk categories to evaluate travel readiness. High-risk cases are verified by a General Physician, who provides the final travel recommendation. PilgrimIQ combines real-time weather forecasts, crowd prediction, waiting-time estimation, terrain analysis, and route optimization to generate a personalized Pilgrim Safety Index (PSI). Based on the user's health profile and travel details, it recommends suitable transportation, accommodation, food and dietary plans, hydration reminders, medicine schedules, packing checklists, safe routes, and personalized pilgrimage timelines. During the journey, the platform provides live weather alerts, AI chatbot assistance, journey tracking, emergency support, nearby healthcare services, and continuous risk monitoring.

Developed using React.js, Node.js, Express.js, MongoDB, Python, AI, ML, NLP, and OCR, PilgrimIQ offers a secure, intelligent, and scalable platform that integrates healthcare support, travel planning, accommodation and food recommendations, and real-time journey assistance to enable safer and more informed pilgrimage experiences across multiple religious destinations.